Optimal mechanism

Optimal mechanism, IPVM

- What's optimal? For the seller, to maximize expected revenue; for society, to maximize social surplus.
- Social surplus: The four classic auctions without reserve price maximize expected social surplus.
- Myerson (1981) identifies mechanism that maximizes seller's expected revenue.
- $X_i = [\underline{x}_i, \overline{x}_i]$; we'll use $\underline{x}_i = 0$ and $\overline{x}_i = 1$; one good, I bidders; $x_i \in X_i$ distributed according to cdf $F_i(x_i)$ with pdf $f_i(x_i)$ such that $(\forall x_i \in X_i)\ f_i(x_i) > 0$.
- See Monteiro and Svaiter (2010) for weaker distributional assumptions.

Optimal mechanism

Objective function

$$\max_{\{p_i,t_i\}_1^I} \quad E\left(\sum_{i=1}^{I} T_i(x_i)\right)$$

$$\forall i, \quad p_i \in L^1([0,1]^I), t_i \in L^1([0,1]^I)$$

$$\forall p_i \geq 0,$$

$$\text{subject to}$$

Optimal mechanism

Constraints

$$(1G)\ \sum_{i=1}^{I} p_i \leq 1$$

$$(IC)\ \forall i\ \forall x_i,\ \ U_i(x_i) \geq U_i(x_i'|x_i)\ \ \forall x_i' \in x_i$$

$$(IR)\ \forall i\ \forall x_i,\ \ U_i(x_i) \geq 0$$

$$E\left(\sum_{i=1}^{n} T_i(x_i)\right) \geq 0.$$

Optimal mechanism

Defining constraints

$$\forall i \; \forall x_i \; \forall x_i',$$

$$\begin{aligned}
Q_i(x_i) &= E_{x_{-i}} p_i(x_{-i}, x_i) \\
T_i(x_i) &= E_{x_{-i}} t_i(x_{-i}, x_i) \\
U_i(x_i' | x_i) &= Q_i(x_i') \cdot x_i - T_i(x_i') \\
U_i(x_i) &= U_i(x_i | x_i)
\end{aligned}$$

The program

$$
\begin{aligned}
&\max_{\{p_i\}_{i=1}^{I}} \quad E\Big[\sum_{i=1}^{n} T_i(x_i)\Big] \\
&s.t. \quad \forall i,\ \forall x = (x_1, \dots, x_I) \\
(1G) \qquad & p_i(x) \geq 0 \ \wedge \sum_{i=1}^{n} p_i(x) \leq 1 \\
(IC) \qquad & Q_i(x_i) = E_{x_{-i}} p_i(x_i, x_{-i}), Q_i \text{ is nondecreasing,} \\
& U_i(x_i) = U_i(0) + \int_0^{x_i} Q_i(y)\,\mathrm{d}y \\
(IR) \qquad & U_i \geq 0.
\end{aligned}
$$

$$U_i(x_i) = Q_i(x_i)x_i - T_i(x_i)$$
$$T_i(x_i) = Q_i(x_i)x_i - U_i(x_i).$$

Revenue in terms of probability

Lemma 1 IPVM, $N = 1$ good. Let (p, t) be a DRM. The seller's expected revenue is

$$\sum_{i=1}^{I} E[T_i(x_i)] = \sum_{i=1}^{I} \int_0^1 \underbrace{\left[x_i - \frac{1 - F_i(x_i)}{f_i(x_i)}\right]}_{v_i(x_i)} Q_i(x_i) f_i(x_i)\, \mathrm{d}x_i - U_i(0).$$

Note

- $v_i(x_i)$ is called *agent i's virtual utility* in Myerson (1981).
- If $X_i = [\underline{x}_i, \overline{x}_i]$, the lower limit of integration is $\underline{x}_i$ instead of zero, and $U_i(0)$ becomes $U_i(\underline{x}_i)$.
- This Lemma also proves the Revenue Equivalence Theorem.

Proof.

$$
\begin{aligned}
E[T_i] &= \int_0^1 [Q_i(x_i) \cdot x_i - U_i(x_i)]\, f_i(x_i)\, \mathrm{d}x_i \\
&= \int_0^1 \left[Q_i(x_i) \cdot x_i - U_i(0) - \int_0^{x_i} Q_i(y)\, \mathrm{d}y\right] f_i(x_i)\, \mathrm{d}x_i \\
&= \int_0^1 (Q_i(x_i) \cdot x_i) f_i(x_i)\, \mathrm{d}x_i - \int_0^1 \int_0^{x_i} Q_i(y)\, \mathrm{d}y\ f_i(x_i)\, \mathrm{d}x_i \\
&\qquad - U_i(0).
\end{aligned}
$$

Note: This step used the integrability condition included in IC but not the convexity of U_i (that is to say, the increasingness of $Q_i(x_i)$).

Integrate the second term by parts using

$$\mathcal{U}(x_i) = \int_0^{x_i} Q_i(y)\,\mathrm{d}y \text{ and } \mathcal{V}'(x_i) = f(x_i)\,\mathrm{d}x_i,$$

$$\int_0^1 \mathcal{U}\mathcal{V}' = \mathcal{U}(1)\mathcal{V}(1) - \mathcal{U}(0)\mathcal{V}(0) - \int_0^1 \mathcal{U}'V$$

$$\begin{aligned}\int_0^1 \left[\int_0^{x_i} Q_i(y)\,\mathrm{d}y\right] f_i(x_i)\,\mathrm{d}x_i &= \int_0^1 Q_i(y)\,\mathrm{d}y - \int_0^1 Q_i(x_i)F_i(x_i)\,\mathrm{d}x_i \\ &= \int_0^1 Q_i(x_i)(1 - F_i(x_i))\,\mathrm{d}x_i.\end{aligned}$$

Replacing

$$
\begin{aligned}
E[T_i] &= \int_0^1 Q_i(x_i) x_i f_i(x_i)\,\mathrm{d}x_i - \int_0^1 Q_i(x_i)(1 - F_i(x_i))\,\mathrm{d}x_i - U_i(0) \\
&= \int_0^1 \underbrace{\left[x_i - \frac{1 - F_i(x_i)}{f_i(x_i)}\right]}_{v_i(x_i)} Q_i(x_i) f_i(x_i)\,\mathrm{d}x_i - U_i(0)
\end{aligned}
$$

QED

Note: $v_i(x_i)$ is called agent i's *virtual utility* (Myerson (1981)).

The program simplified

$$\max_{\{p_i\}_{i=1}^{I}} \sum_{i=1}^{I} \int_0^1 \overbrace{\left[x_i - \frac{1-F_i(x_i)}{f_i(x_i)}\right]}^{v_i(x_i)} Q_i(x_i) f_i(x_i)\,\mathrm{d}x_i - U_i(0)$$

$$s.t. \quad (\forall i)\ (\forall x = (x_1, \ldots, x_I))$$

$$(1G) \quad p_i(x) \geq 0 \ \wedge \sum_{i=1}^{I} p_i(x) \leq 1$$

$$(IC) \quad Q_i \text{ nondecreasing, } U_i(x_i) = U_i(0) + \int_0^{x_i} Q_i(y)\,\mathrm{d}y$$

$$(IR) \quad U_i \geq 0.$$

The final program

$$\max_{\{p_i\}_{i=1}^I} \sum_{i=1}^{I} \int_0^1 \overbrace{\left[x_i - \frac{1-F_i(x_i)}{f_i(x_i)}\right]}^{v_i(x_i)} Q_i(x_i) f_i(x_i)\,\mathrm{d}x_i$$

$$s.t. \quad (\forall i)\ (\forall x = (x_1, \ldots, x_I))$$

$$(1G) \quad p_i(x) \geq 0\ \wedge \sum_{i=1}^{I} p_i(x) \leq 1$$

$$(IC) \quad Q_i \text{ nondecreasing, } U_i(x_i) = U_i(0) + \int_0^{x_i} Q_i(y)\,\mathrm{d}y$$

where $Q_i(x_i) = E_{x_{-i}} p_i(x_i, x_{-i})$

The optimal mechanism

$$\max_p \sum_{i=1}^{I} \int_0^1 v_i(x_i) \overbrace{Q_i(x_i)}^{E_{x_{-i}}[p_i(x_i, x_{-i})]} f_i(x_i)\,\mathrm{d}x_i$$

$$v_i(x_i) = \left[x_i - \frac{1 - F_i(x_i)}{f_i(x_i)} \right].$$

Regularity condition: $(\forall i)\ v_i(x_i)$ is strictly increasing.

Theorem 1 (Myerson) IPVM and the regularity condition holds. The optimal DRM is

$$p_i(x_i, x_{-i}) = \begin{cases} 1 & \text{if } v_i(x_i) > \max\left\{0, \{v_j(x_j)\}_{j \neq i}\right\} \\ 0 & \text{otherwise} \end{cases}$$

Interpretation.

- The object is sold to the bidder with highest $v_i(x_i)$ provided it is nonnegative.
- If v_i is strictly increasing and symmetry is assumed, the bidder with highest v_i is the bidder with the highest valuation.

Proof (Myerson). By *observation.*

$$\max_p \sum_{i=1}^{I} \int_0^1 v_i(x_i) Q_i(x_i) f_i(x_i) \, \mathrm{d}x_i$$

$$\max_p \sum_{i=1}^{I} \int_0^1 \dots \int_0^1 v_i(x_i) p_i(x_i, x_{-i}) f_1(x_1) \cdots f_I(x_I) \, \mathrm{d}x_1 \cdots \mathrm{d}x_I$$

- Propose a mechanism $\{p_i\}_{i=1}^{I}$ that maximizes the OF without considering the IC constraint. Then verify that the proposed mechanism satisfies IC.

It remains to show that the proposed solution satisfies the IC constraint.

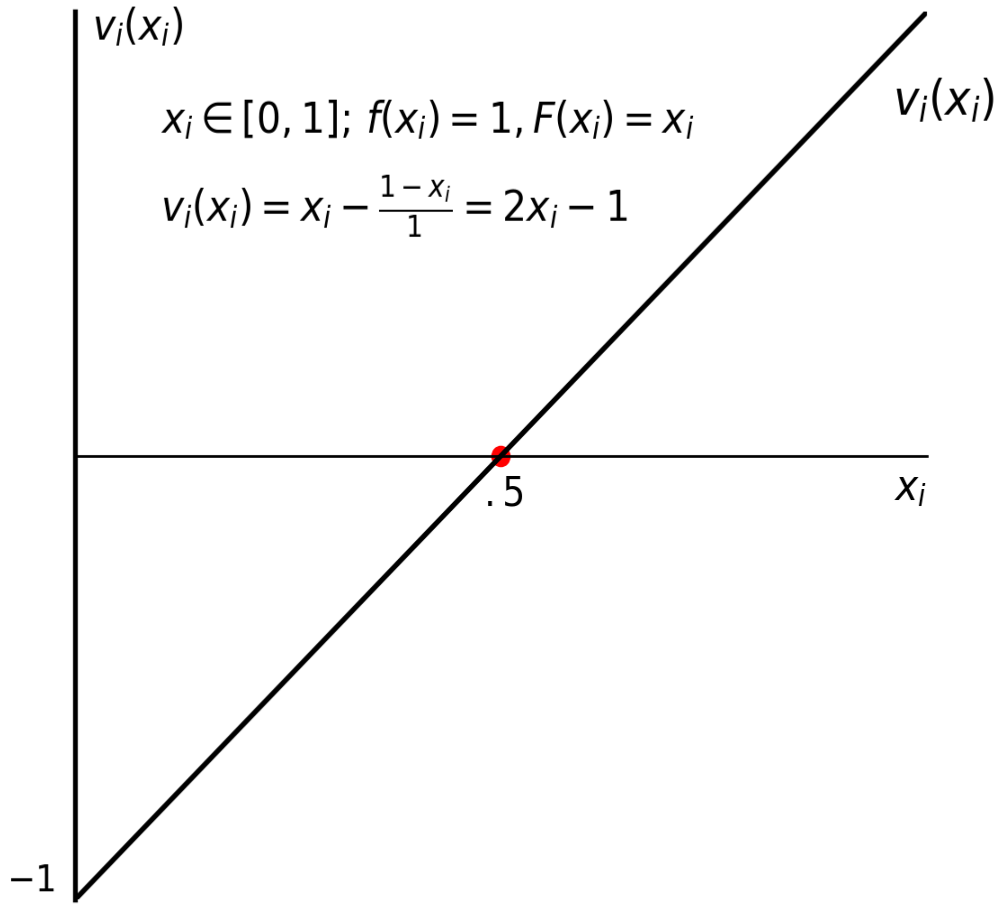
- Since $v_i(x_i)$ is strictly increasing, $x_i' > x_i \implies v_i(x_i') > v_i(x_i)$.
- By definition of the proposed p_i,

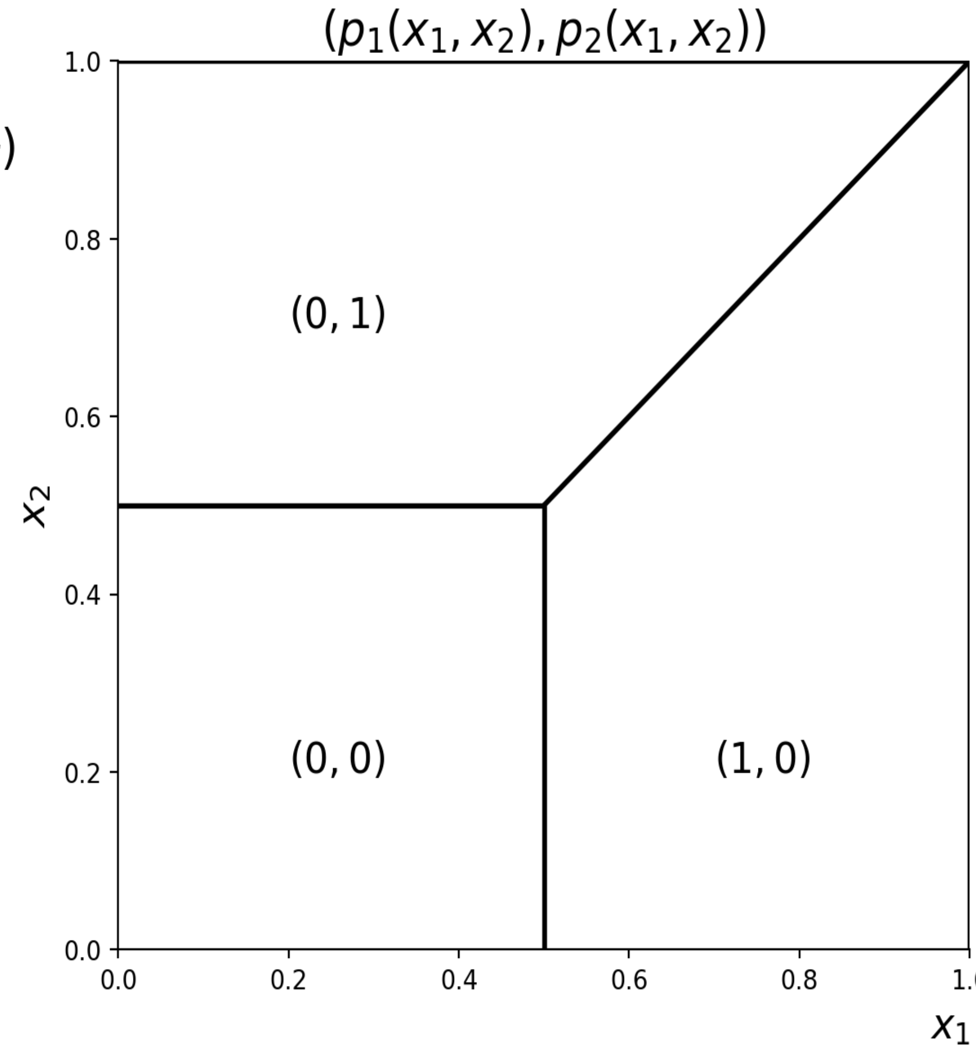
$$(\forall x_{-i})\ p_i(x_i', x_{-i}) \geq p_i(x_i, x_{-i}).$$

- $E_{x_{-i}} p_i(x_i', x_{-i}) \geq E_{x_{-i}} p_i(x_i, x_{-i})$.

This proves $\{p_i\}_{i=1}^I$ satisfies IC and completes the proof. QED

Example: regularity and symmetry





Remarks, regularity and symmetry in IPVM.

- Institution implementing optimal DRM is any of the classic auctions with reserve price.
- The reserve price, which is $r = v^{-1}(0) = 0.5$ in the uniform case, does not change with the number of bidders I.
  - Reserve price is only useful if $I - 1$ bidders have *low* valuations and the winner has a *high* valuation.
- If $x_i = [\underline{x}_i, \bar{x}_i]$, then for large $\underline{x}_i$, the optimal reserve price is zero.
- $I = 1$ leads to a bargaining or monopoly interpretation. The optimal strategy is a take-it-or-leave-it offer r; set price $price = r$.

- With perfect information, mechanism that maximizes seller's revenue is efficient.

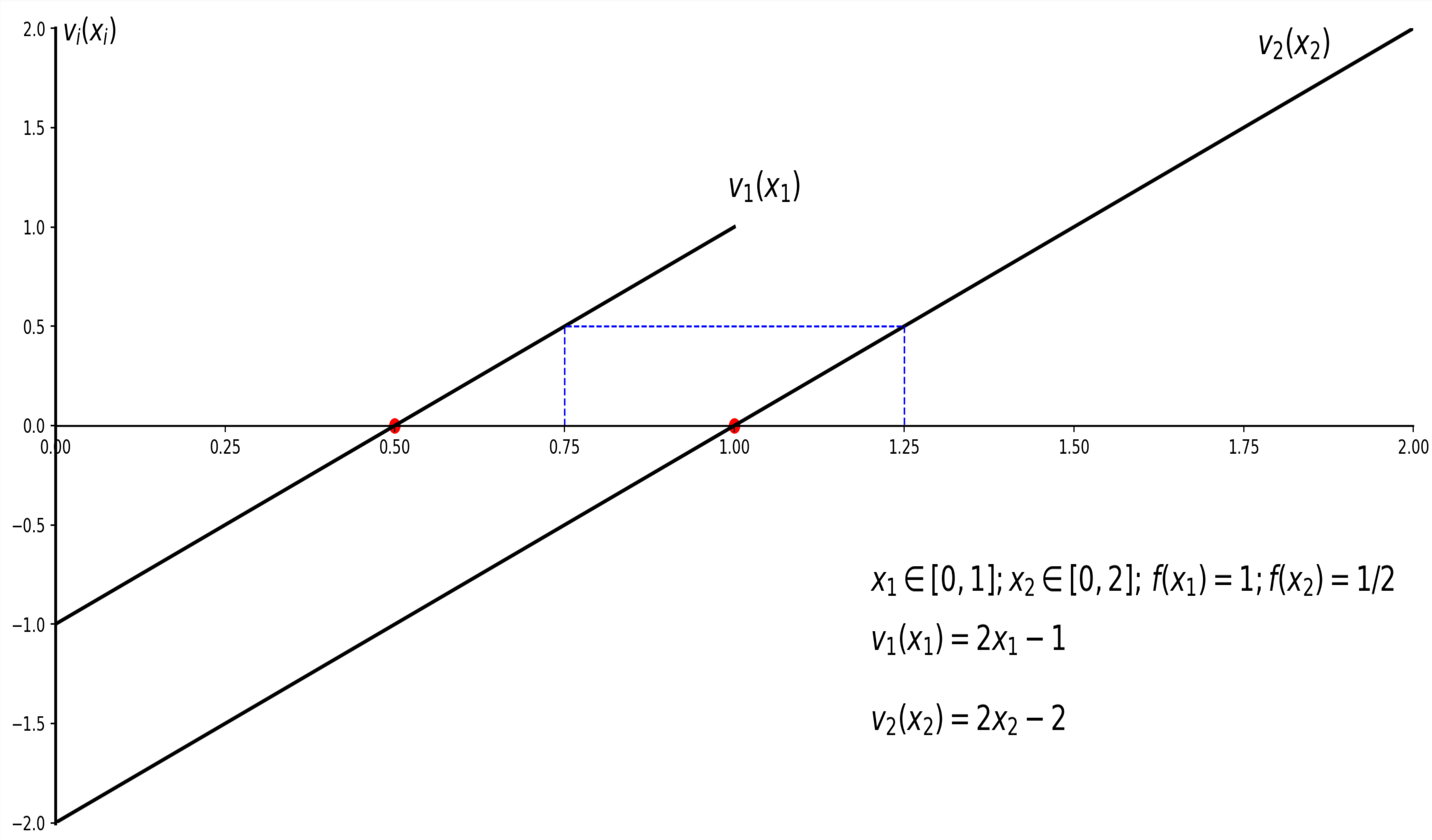
  Observation: the perfectly discriminating monopolist maximizes revenue by making the pie as large a possible and then eating it all.

- With incomplete information, seller's optimal mechanism implies seller keeps object with certain probability, in the example that probability is $\left(\frac{1}{2}\right)^I$.
- Optimal social mechanism: any of the classic auctions without reserve price.
- If in addition the seller's valuation is private information, is the socially optimal mechanism efficient?

Example non-symmetric IPVM

Without symmetry, the optimal DRM may assign object to bidder that does not have the highest valuation.

- $i = 1, 2$, $x_i \in [0, \bar{x}_i]$, $\bar{x}_1 = 1$, $\bar{x}_2 = 2$. Uniform distribution.
- $v_i(x_i) = 2x_i - \bar{x}_i$.
- $v_1(x_1) = 2x_1 - 1$.
- $v_2(x_2) = 2x_2 - 2$.
- $v_1^{-1}(0) = r_1 = 0.5$, and $v_2^{-1}(0) = r_2 = 1$.



$v_i(x_i)$
$v_2(x_2)$
$v_1(x_1)$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
$x_1 \in [0,1]; x_2 \in [0,2];\ f(x_1) = 1; f(x_2) = 1/2$
$v_1(x_1) = 2x_1 - 1$
$v_2(x_2) = 2x_2 - 2$

- Relatively less work on asymmetric environments.
- In example, bidder 1 may win the object even if bidder 2 has the highest valuation.

  The optimal auction *discriminates* against bidders with higher up potential, i.e. high x_i. In the DRM this discrimination discourages those bidders from under representing values close to their upper limit.
- Distinction between DRM and implementing institution.
- See Manelli and Vincent (1995), Manelli and Vincent (2004), Zhang (2026) in the context of procurement auctions.

References

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Zhang, Kun. 2026. “Optimal Procurement Designed: A Reduced Form Approach.” *American Economic Journal, Microeconomics*.